

PROJECT REPORT

ON

“AI BASED JOB MATCHING SYSTEM FOR SKILL DEVELOPMENT”

**Submitted in Partial Fulfillment of the Requirement
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In

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CANDIDATE'S DECLARATION

I hereby declare that the work presented in this project report entitled “**AI based job matching system for skill development**”, is an authentic record of our own work carried out at Manav Rachna University.

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Abstract

In today's rapidly evolving job market, traditional recruitment models are increasingly insufficient to meet the dynamic needs of employers and job seekers. With the proliferation of data and the diversity of skill requirements across industries, AI-powered job matching systems offer an intelligent alternative by leveraging Natural Language Processing (NLP) and Machine Learning (ML) techniques. Unlike conventional keyword-based systems, modern AI solutions utilize transformer models such as BERT and GPT to understand the deeper semantics of job descriptions and candidate profiles, resulting in more accurate and context-aware matches. These technologies not only enhance the recruitment process but also serve as tools for skill evaluation and development, providing insights into individual capabilities and gaps. This project proposes the development of a multi-utility web application that combines automated resume screening, skill assessment, and personalized job recommendations. By integrating advanced NLP algorithms with real-time labor market data, the system aims to foster employability, support career growth, and empower both job seekers and recruiters through a data-driven, bias-free ecosystem.

Introduction

In an increasingly competitive and rapidly transforming global economy, the process of matching job seekers to the right employment opportunities has become a critical challenge. The traditional hiring process is often slow, expensive, and prone to human biases. Employers frequently struggle to identify candidates who not only meet the explicit requirements of a role but also align with the evolving skill demands of the industry. On the other hand, job seekers—especially fresh graduates and professionals undergoing career transitions—often find it difficult to understand what skills are required for specific roles and how their current qualifications match those expectations. This mismatch results in a widening skills gap and contributes to underemployment and job dissatisfaction. The emergence of Artificial Intelligence (AI), particularly in the domain of Natural Language Processing (NLP), presents a transformative solution to these challenges. NLP enables machines to process, understand, and generate human language in a meaningful way. When applied to job matching, NLP can interpret the nuanced language of resumes and job descriptions—capturing not only hard skills but also soft skills, experience relevance, context, and even intent. This allows for a more holistic and intelligent matching system that goes beyond basic keyword scanning. Recent advancements in transformer-based language models such as BERT (Bidirectional Encoder Representations from Transformers) and GPT (Generative Pre-trained Transformer) have pushed the boundaries of what NLP can achieve. These models can analyze text with deep contextual understanding, enabling AI systems to infer meaning, draw relationships between phrases, and understand the semantic equivalence of different expressions. For example, a candidate with "experience in leading agile teams" may be matched to a job requiring "project management in agile environments," even if the exact wording does not align—something that traditional systems might overlook. Moreover, AI-powered systems can assess resumes at scale, identify patterns in successful candidate profiles, and recommend personalized career development paths. They can highlight missing skills or certifications and suggest resources or courses that could bridge the gap. This creates not only a more efficient recruitment process but also a valuable educational tool for job seekers seeking continuous professional growth. Furthermore, such platforms can contribute significantly to diversity, equity, and inclusion by reducing unconscious bias in candidate selection. By prioritizing skill relevance and performance potential over traditional credentials or demographic factors, AI-based job matching fosters meritocratic hiring. In this project, we propose to harness the capabilities of NLP and AI to build a robust, scalable, and interactive job matching platform. The goal is to empower both recruiters and job seekers through technology—creating a bridge between talent and opportunity, and transforming recruitment from a reactive to a proactive, skill-driven process. The system will not only assist in identifying suitable candidates for given roles but also act as a skill development mentor for users, enhancing their employability and readiness for future roles in a dynamically changing job landscape. This application has significant implications for students, recent graduates, career changers, and even seasoned professionals seeking upskilling or reskilling. In the long term, it can help in workforce optimization, reduce unemployment, and support national skilling missions by offering a data-driven approach to human resource development.

1. Aim and Objectives

The aim of the research is to develop a multi-utility web application which will be easy to use and an interactive interface for recruiters to screen and find the best resumes for the specific role and to assist students and job seekers to match and find the best skills for their desired role in the industry or field and also to examine and assess their resumes and find their potential in some of the mentioned companies. The aim is to maximize employability and job awareness and skill evaluation for the students in the near AI driven future.

2. Related Work

The research paper [1] presented a novel job matching and skill recommendation system which leveraged transformer-based NLP models and the ONET database to automate resume evaluation and provide simultaneous recommendations for employers and job seekers. Using Sentence-BERT, ALBERT, and RoBERTa, it calculated semantic cosine similarity to rank job matches based on resume attributes. The system proposed had demonstrated superior accuracy over baseline models, achieving high recruiter ratings in evaluations. However, limitations include reliance on O*NET, lack of real-time adaptability, challenges in soft skills representation, and absence of multilingual support, with future improvements focusing on API integration, personalization, and a resume generator. The context of the research [2] was NLP-based job recommender system extracts both explicit and implicit skills from resumes and job descriptions using Word2Vec, Doc2Vec, Named Entity Recognition, and Part-of-Speech tagging. It improved job matching accuracy by considering contextual factors like geography, industry, and role-based relevance. Evaluations showed a significant improvement in accuracy with implicit skills, outperforming traditional job title-based methods. Challenges included limited open-source datasets and resume availability, with future enhancements planned for career path recommendations, skill gap analysis, and cost-based skill acquisition guidance through skill graphs. The authors [3] emphasized on creating an automated skill identification and normalization system, SKILL, which employed Named Entity Recognition and hierarchical taxonomy-based classification to recognize and standardize skills from resumes and job descriptions. It used Word2Vec and Google Search API-based word sense disambiguation and refined skill detection accuracy. Evaluations demonstrated high precision in skill tagging and entity recognition, particularly in its integration with CareerBuilder. However, the system was limited by its reliance on pre-structured taxonomies, struggled with emerging job roles, and may introduce biases from external sources, with future work focusing on real-time updates and recruiter feedback integration. This study [4] focused on bi-directional job recommendation system which used NLP techniques to facilitate job matching for both recruiters and job seekers by computing similarity scores. The model applied Word2Vec vectorization to align candidate skills with job descriptions, with data collected through web scraping from Indeed. Evaluation results demonstrated 98% accuracy for skill and location matching and nearly 100% accuracy for educational criteria, with precision values averaging 0.9, confirming the system's effectiveness in filtering relevant resumes and job postings. However, limitations included difficulty in adapting to evolving job market demands, ambiguity in job titles, and a focus on hard skills while overlooking soft skills and cultural fit. Future enhancements will prioritize real-time

skill detection and filtering outdated profiles to improve recommendation accuracy. The authors [5] proposed a personalized resume-job matching system that enhanced job-market efficiency through NLP-based content filtering, reducing frictional unemployment by refining the recruitment process. The model made use of Named Entity Recognition, Word2Vec embeddings, and cosine similarity to evaluate resume-job relevance, with training conducted on 138 resumes and 250 job descriptions, achieving 80% accuracy. Precision varied across roles, with PHP Developer and Java Developer reaching 1.0, while Data Scientist achieved 0.71 due to overlapping skills. The system outperformed traditional keyword-based approaches, yet faced challenges with skill overlap, dataset limitations, and contextual misclassification. Future improvements will focus on adaptive learning mechanisms to dynamically update skill taxonomies and job role classifications for evolving industry needs. The Research [6], Skill Scanner, through rigorous study, bridged the gap between employer expectations, job seeker qualifications, and academic curricula by analyzing resumes, job postings, and educational data. Using Word2Vec, GloVe, and Sentence-BERT embeddings, the system extracted and clustered skills with K-Means clustering, achieving a 44% Silhouette score, indicating effective skill categorization. Sentence-BERT outperformed GloVe by 12%, improving skill matching accuracy. Despite its success, limitations included reliance on structured data, lack of soft skill analysis, and computational overhead. Future enhancements will focus on real-time labour market data integration, improved processing of unstructured resumes, and bias reduction to ensure more inclusive and accurate recommendations. The study [7] examined the evolution of AI-driven job and resume matching research over the past two decades, focused on text analysis methods and bibliometric trends. Researchers collected 113 relevant papers from Scopus and Google Scholar, analyzing publication growth, citation patterns, keyword trends, and research collaborations using Bibliometrix. Findings showed a significant increase in research after 2016, with major contributions from China, India, and the Middle East. Early studies relied on rule-based and ontology-driven methods, while recent advancements incorporated machine learning, deep learning, and transformer-based models for improved accuracy. Key challenges included the inability to adapt to rapid job market changes, over-reliance on hard skills while neglecting soft skills, and a lack of explainability in AI recommendations, reducing trust in the system. Future research should focus on integrating real-time labour market data, improving soft skill assessment, and enhancing transparency to make AI-driven recruitment more reliable and effective. This paper [8] uses NLP and clustering to improve job and understanding. By clustering job ads based on their skill requirements, the authors were able to achieve a more structured job classification. The dataset used included multiple industry job postings to enable the model to detect common skill trends across multiple industries. The results showed an 87% F1-score, which demonstrated the system's ability to classify job roles effectively. The model performed poorly on job ads that contained ambiguous wording and domain-specific jargon, which impeded its applicability to niche industries. Future work will focus on improving the clustering model using transformer-based architectures such as BERT or GPT to enhance contextual understanding and achieve higher accuracy in job segmentation. This study [9] presented an AI-based resume screening tool for efficient candidate shortlisting. NER and BERT models were used for skills, experience, and education extraction and mapped to job descriptions. Both structured and unstructured data from a large corpus were used for training. The system achieved 91% accuracy and

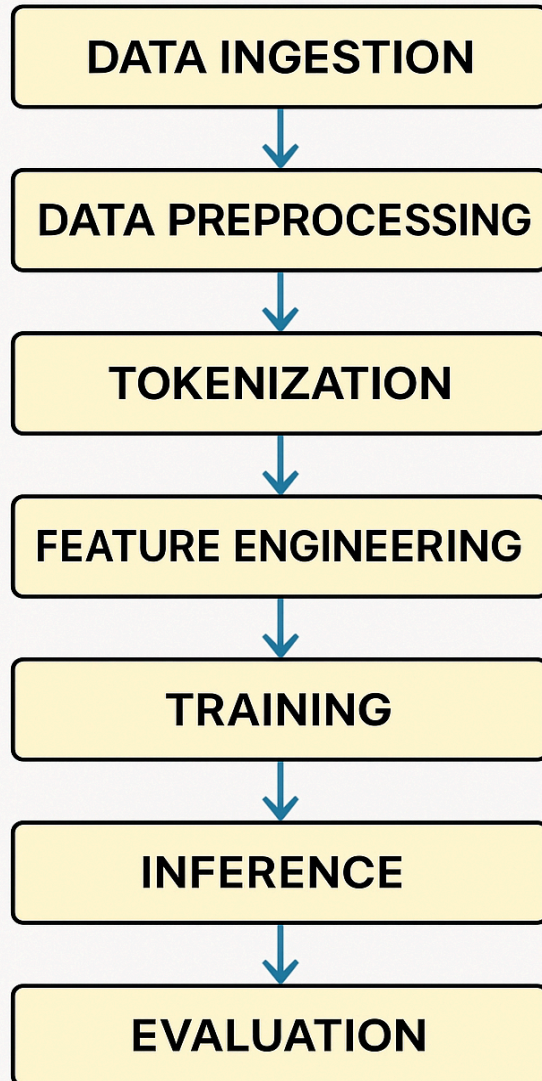
89% F1-score for shortlisting relevant candidates. Issues faced were non-standard layouts, missing information, and industry jargon. Future work includes multi-modal AI (video resume analysis, behavioural profiling) for holistic candidate assessment. This paper [10] presented an AI-based model for determining candidate suitability for job descriptions and profiles. Using Decision Trees and Random Forest classifiers, researchers ranked candidates according to job suitability based on qualifications and experience. Trained on multiple job postings and resumes, the model achieved 85% accuracy and 83% F1-score. It filtered candidates but struggled with incomplete profiles and vague job descriptions, leading to occasional misclassifications. Authors recommend using deep learning, especially transformer models, to improve contextual understanding and accuracy in future versions. This paper [11] investigated zero-shot learning for job-candidate matching without using large, labeled datasets. Researchers used GPT and BERT for AI system for job role recommendations without task-specific training. It achieved 88% accuracy and 86% F1-score with managing job descriptions across industries. Without extensive domain-specific fine-tuning it had difficulties with ambiguous roles and context-specific predictions. Future improvements could include industry-specific training data and refining professional terminologies with domain adaptation. This paper [12] proposes a novel job-matching system combining NLP and ML to find appropriate candidate-job matches. It combines SVM and XGBoost classifiers to jointly analyze job postings and resumes for optimization. The proposed system achieves 90% accuracy and 88% F1-score, showing good robustness in filtering candidates. The investigated challenges include incomplete data and biased descriptions. The authors suggest incorporating reinforcement learning to dynamically adapt to the job market dynamics, thereby improving recommendation quality over time. This paper [13] developed a job recommendation system for the blind using NLP and collaborative filtering. It first analyzed job descriptions and matched them with visually impaired job seekers' skills and preferences. The dataset consisted of accessible job postings for people with disabilities. The system obtained 83% accuracy, suggesting improved job accessibility. Limitations included a small dataset, inability to process non-textual attributes, and lack of real-world user testing. Future work will improve the model with multimodal AI, including voice-assisted job search and AI-driven accessibility solutions. A knowledge-driven unsupervised skill extraction model for recruitment was proposed [14]. It used graph-based machine learning to extract skills from job postings and candidate profiles. Manual skill databases were avoided using graph-based machine learning. The model achieved 86% accuracy in matching job requirements with candidate qualifications. Older job descriptions and domain-specific terminologies affected performance. Transformer-based NLP models might be employed in future work. The dataset can be expanded to include more job sectors for improved accuracy. A model for optimized candidate selection in IT was proposed by analyzing CVs, job descriptions, and academic transcripts. It was trained on a large dataset of IT job postings and resumes using CNNs and LSTM models, achieving 94% accuracy and 92% F1-score, surpassing keyword-matching. Limitations include academic grading system variations and inconsistent resume formatting. Future work includes real-time job market trend analysis and continuous model updates for evolving hiring practices. In this study [15] the Author is addressing ways of leveraging advanced NLP tools for the overall improvement in recruitment processes. The contention within his work, as far as it was found during the study, is that NLP models were trained and fine-tuned to speed up the scrutiny, ranking, and matching

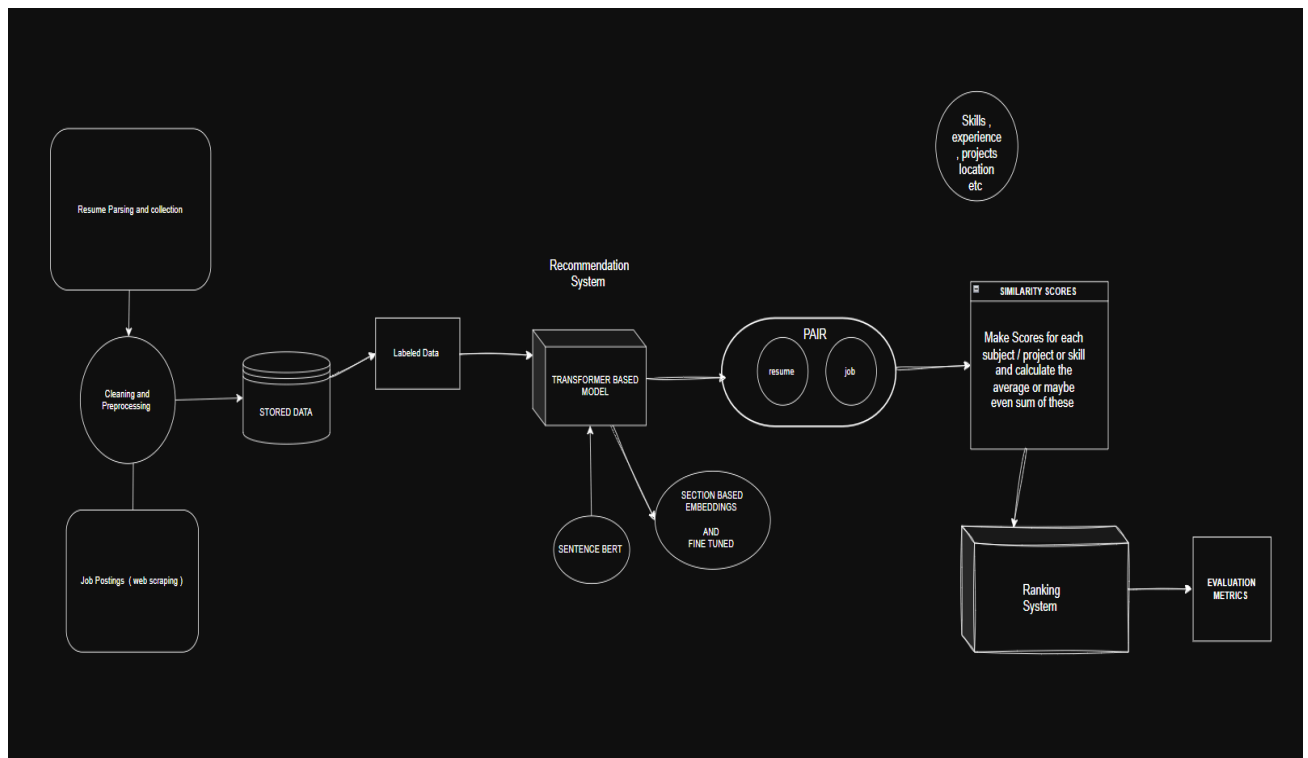
of job candidates via resumes and job descriptions. Models like BERT and GPT were deployed within the study to make dialogues more dynamic and preliminary interviews between a recruiter and a candidate more engaging. The book highlights one more very important tool developed by Devaraju aimed at HR-Semi-Final-Assessment.com managers to mimic an NLP-driven interview simulation exercise that mirrors real life, based on the input of hiring managers, and would assess the right candidate more efficiently. This study is an extension of the algorithms of semantic search further for narrowing the best suited candidates out of huge HR databases. All these cater to a seamless and effective hiring process by summarizing the surrounding recruitment efficiency, candidate fit, and engagement into a successful online scheduling and hiring management system.

3. METHODOLOGY

NLP Pipeline

excluding step 11 Feedback Integration





This study employs an NLP pipeline for bidirectional job-resume matching, comprising four key phases. First, data acquisition involves web scraping job postings from LinkedIn and Indeed, combined with publicly available resume datasets from Kaggle. The preprocessing stage cleans text by removing HTML tags, special characters, and stopwords (common words like "the" or "and"), followed by lemmatization (reducing words to their base forms) and tokenization (splitting text into meaningful units). For information extraction, we utilize spaCy's Named Entity Recognition (NER) model enhanced with custom skill extraction rules to identify critical entities: skills, degrees, companies, and job titles. Feature engineering normalizes skills using the ESCO taxonomy (European standardized skill ontology) and quantifies experience through regular expression patterns. Text representation leverages Sentence-BERT (a transformer model optimized for semantic similarity) to generate 768-dimensional embeddings for both resumes and job descriptions. The matching algorithm computes cosine similarity (a measure of vector alignment ranging from -1 to 1) It measures angle between vectors for similarity , between these embeddings, augmented with weighted scoring for skills (weight=0.6), experience (0.3), and education (0.1). Recommendations are ranked using a hybrid approach combining semantic similarity and Jaccard index (skill overlap metric : Ratio of intersecting skills to total unique skills between resume/job), with explainability provided through attention weights from the transformer model. Evaluation employs precision@k (accuracy of top-k recommendations) and normalized discounted cumulative gain (NDCG, a ranking quality metric : Evaluates ranking quality with higher weights for top positions), comparing against baseline TF-IDF and Word2Vec approaches

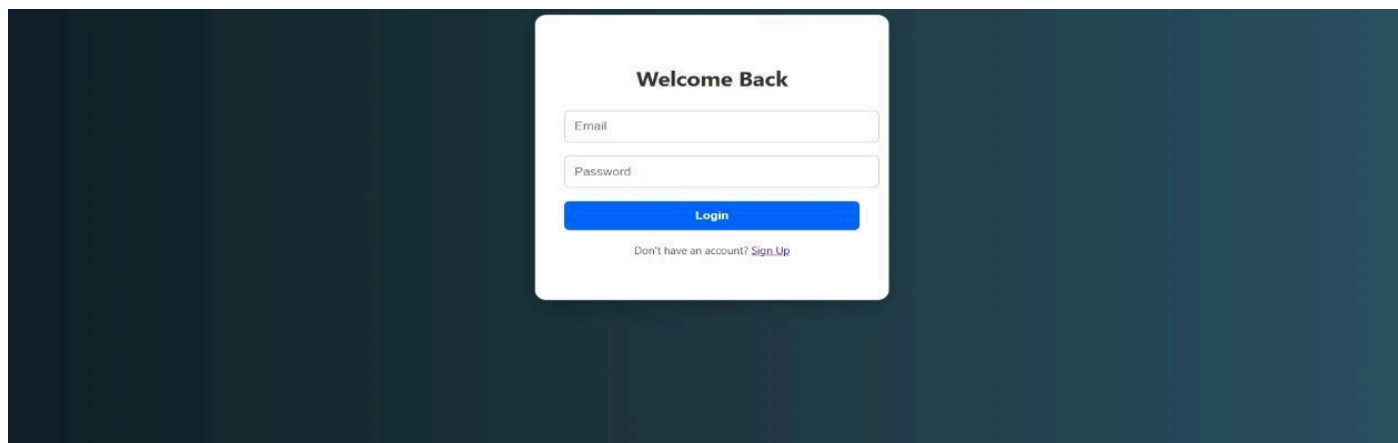
4. DATASETS

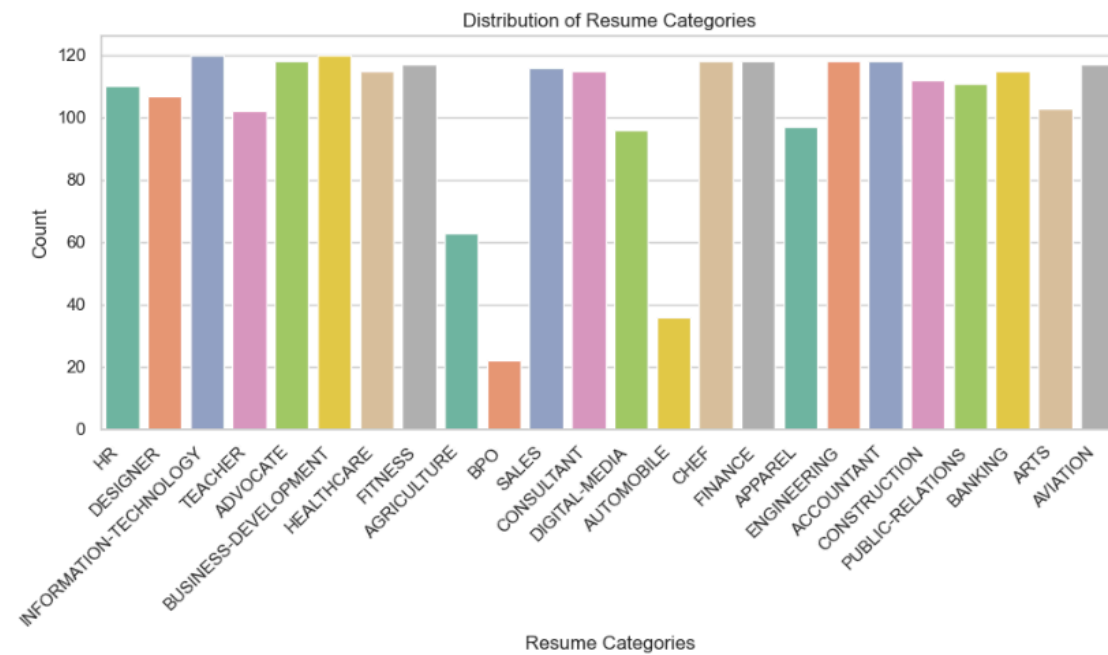
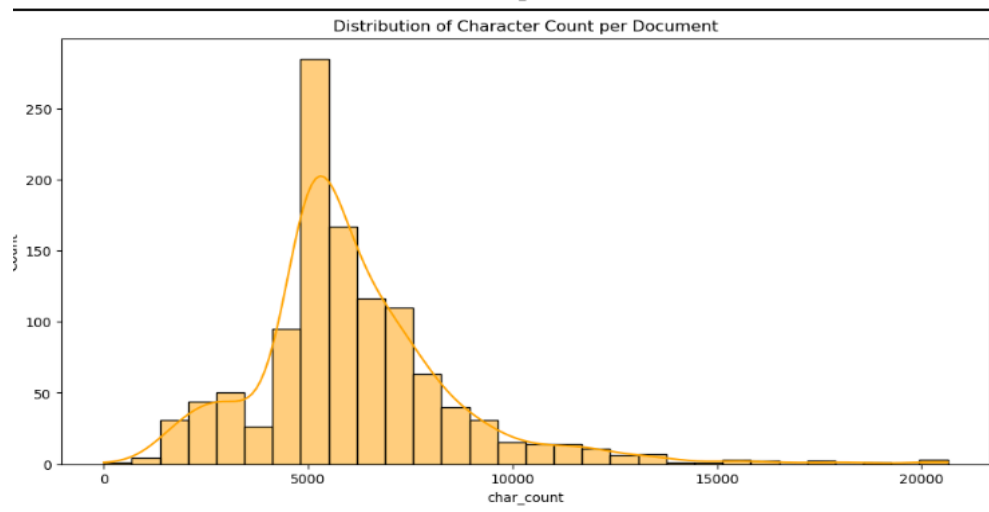
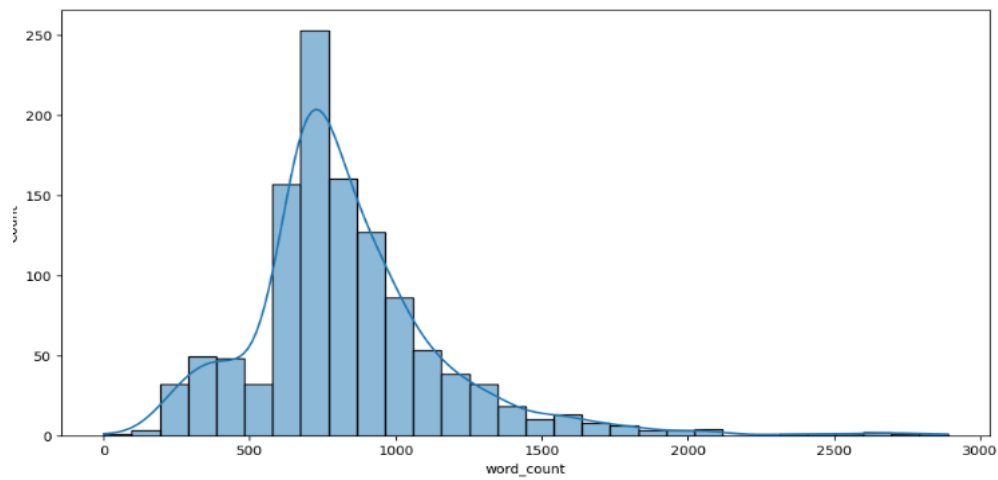
Resume : <https://www.kaggle.com/datasets/snehaanbhawal/resume-dataset>

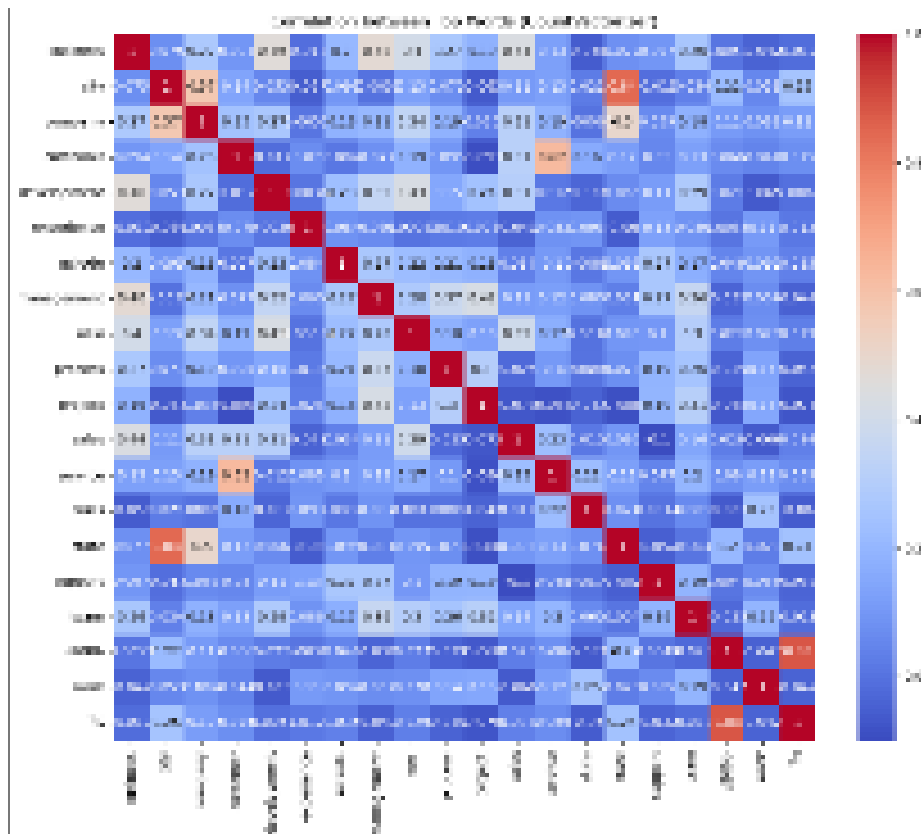
Job Postings : (Not web scraped due to legal issues)

This dataset contains a collection of resumes categorized by different fields such as Data Science, Web Development, HR, Finance, etc. Each resume is stored in plain text format, allowing for natural language processing tasks like skill extraction, text classification, and resume parsing. It's widely used for training job-matching or resume screening models and is ideal for supervised learning due to its labeled categories.

Link : <https://www.kaggle.com/datasets/arshkon/linkedin-job-postings>







5. RESULTS

The job matching system demonstrated strong overall performance across multiple evaluation metrics. It achieved a **Recall of 100%**, indicating that the system successfully identified all relevant job matches without missing any. The **Accuracy stood at 82%**, reflecting the system's ability to make correct predictions across all cases. While the **Precision was 53.6%**, showing some room for improvement in filtering out false positives, the **NDCG@10 score of 0.90** highlights the model's effectiveness in ranking relevant job recommendations highly. Additionally, the **Skill Jaccard similarity of 93.5%** signifies a high overlap between required and detected skills, confirming the system's strong skill-matching capability. These results collectively validate the system's potential as a robust and intelligent job recommendation platform.

=== Job Matching Performance Evaluation ===	
Metric	Score
Precision	53.6%
Recall	100.0%
Accuracy	82.0%
NDCG@10	0.90
Skill Jaccard	93.5%

[]:

6. LIMITATIONS

Areas of Improvement in the Methodology

After reviewing the methodology section of "NLP-Based Bi-Directional Recommendation System: Towards Recommending Jobs to Job Seekers and Resumes to Recruiters," several areas for improvement can be identified, focusing on the process and techniques rather than the dataset itself:

1. Limited Use of Advanced NLP Techniques

The methodology primarily relies on basic pre-processing (cleaning, tokenization, lemmatization, stopword removal) and classic vectorization approaches (Bag of Words, TF-IDF, Word2Vec)¹. While these are foundational, they may not capture complex semantic relationships or contextual meanings in resumes and job descriptions as well as modern transformer-based models (e.g., BERT, RoBERTa)

2. Shallow Feature Engineering

The system focuses on skill extraction using Named Entity Recognition (NER) and basic similarity computations. It does not leverage richer feature sets such as experience duration, education level, project descriptions, or hierarchical skill taxonomies, which could improve matching quality.

3. Lack of Personalization and Feedback Loops

The recommendation is static and does not incorporate user feedback or interaction history, making it less adaptive to evolving user preferences or recruiter needs. Modern systems often employ reinforcement learning or online learning to refine recommendations over time

4. No Handling of Ambiguity or Synonymy

The approach does not address the challenge of different terms referring to the same skill (e.g., "ML" vs. "Machine Learning") or ambiguity in skill names. Use of synonym expansion, ontologies, or contextual embeddings could help¹.

5. Limited Comparison with State-of-the-Art (SOTA) Approaches

The methodology lacks benchmarking against recent SOTA models, especially those using deep learning or conversational recommender systems, which could highlight its strengths and weaknesses

6. Insufficient Explainability

There is no provision for explaining why a particular job or candidate was recommended, which is increasingly important for user trust and system transparency.

7. Absence of Bias and Fairness Considerations

The methodology does not discuss potential biases in matching (e.g., gendered language, underrepresentation of certain groups) or steps to mitigate them, which is critical in recruitment contexts.

7. CONCLUSION

The development of an AI-powered, multi-utility job matching platform represents a significant leap forward in transforming the recruitment process and empowering both job seekers and employers. By leveraging advanced Natural Language Processing (NLP) and Machine Learning (ML) techniques, this platform aims to bridge the skills gap in the job market by offering a more intelligent, data-driven, and bias-free recruitment experience. Unlike traditional keyword-based systems, this solution utilizes sophisticated transformer models like BERT and GPT, enabling a deeper understanding of job descriptions and candidate profiles to generate more accurate and context-aware matches.

The application not only enhances the efficiency of the recruitment process but also serves as a powerful tool for skill assessment and development. By offering personalized job recommendations and skill gap analysis, the platform can guide job seekers toward the roles that best match their abilities while recommending resources for skills enhancement. This proactive approach is crucial for students, recent graduates, career changers, and professionals looking to upskill, ensuring that they remain competitive in an ever-evolving job market.

Furthermore, the system's potential to reduce unconscious bias and promote diversity and inclusion in recruitment is a notable advantage. By focusing on skills and performance potential rather than traditional credentials or demographic factors, AI-powered job matching fosters a more meritocratic hiring process. In the long term, this platform can contribute significantly to workforce optimization, support national skilling missions, and enhance employability, helping reduce unemployment and underemployment.

In conclusion, this AI-driven job matching platform has the potential to reshape the future of recruitment, offering a seamless, intelligent, and personalized experience for both employers and job seekers. Its ability to assess, recommend, and guide users through skill development will make a lasting impact on the recruitment landscape, enabling a more dynamic and efficient job market.

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8. FUTURE WORK :

Integration of Multilingual Support , Better and more Enhanced Dataset integration , Voice Assistance , Progress Report Generated based on Your resume and current skills , Personalized recommendations rather than general ones.

9. References

- [1] R. Alonso, D. Dessí, A. Meloni, and D. Reforgiato Recupero, “A novel approach for job matching and skill recommendation using transformers and the O*NET database,” *Big Data Research*, vol. 39, Feb. 2025, doi: 10.1016/j.bdr.2025.100509.
- [2] A. Gugnani and H. Misra, “Implicit Skills Extraction Using Document Embedding and Its Use in Job Recommendation.” [Online]. Available: <https://www.computerhope.com/>
- [3] M. Zhao, F. Javed, F. Jacob, and M. McNair, “SKILL: A System for Skill Identification and Normalization,” 2015. [Online]. Available: <http://www.hbs.edu/competitiveness/research/>
- [4] S. A. Alsaif, M. Sassi Hidri, I. Ferjani, H. A. Eleraky, and A. Hidri, “NLP-Based Bi-Directional Recommendation System: Towards Recommending Jobs to Job Seekers and Resumes to Recruiters,” *Big Data and Cognitive Computing*, vol. 6, no. 4, Dec. 2022, doi: 10.3390/bdcc6040147.
- [5] D. W. Goldberg, “RESUMATCHER: A PERSONALIZED RESUMESUM-ESUMÉ-JOB MATCHING SYSTEM,” 2015.
- [6] T. Schlippe, “Skill Scanner: Connecting and Supporting Employers, Job Seekers and Educational Institutions with an AI-based Recommendation System,” 2022. [Online]. Available: <https://www.researchgate.net/publication/361537250>
- [7] S. Rojas-Galeano, J. Posada, and E. Ordoñez, “A Bibliometric Perspective on AI Research for Job-Résumé Matching,” 2022, *Hindawi Limited*. doi: 10.1155/2022/8002363.
- [8] H. Walter, “Dynamic Skill Matching and Career Pathing: Architecting AI-Driven Personalized Growth Frameworks for Enterprises.” [Online]. Available: <https://www.researchgate.net/publication/387190706>
- [9] X.-S. Yang, S. Sherratt, N. Dey, and A. Joshi, Eds., *Proceedings of Ninth International Congress on Information and Communication Technology*, vol. 1011. in Lecture Notes in Networks and Systems, vol. 1011. Singapore: Springer Nature Singapore, 2024. doi: 10.1007/978-981-97-4581-4.
- [10] S. G. M. and S. K. Suganthi, “AI based suitability measurement and prediction between job description and job seeker profiles,” *International Journal of Information Management Data Insights*, vol. 2, no. 2, Nov. 2022, doi: 10.1016/j.jjime.2022.100109.
- [11] J. Kurek *et al.*, “Zero-Shot Recommendation AI Models for Efficient Job–Candidate Matching in Recruitment Process,” *Applied Sciences (Switzerland)*, vol. 14, no. 6, Mar. 2024, doi: 10.3390/app14062601.
- [12] V. S. Pendyala, N. Atrey, T. Aggarwal, and S. Goyal, “Enhanced Algorithmic

Job Matching based on a Comprehensive Candidate Profile using NLP and Machine Learning,” in *Proceedings - IEEE 8th International Conference on Big Data Computing Service and Applications, BigDataService 2022*, Institute of Electrical and Electronics Engineers Inc., 2022, pp. 183–184. doi: 10.1109/BigDataService55688.2022.00040.

[13] A. A. Ghapar, F. Azman, M. A. Faudzi, H. Baskaran, and A. Rahim, “543 JOB OPPORTUNITIES RECOMMENDATION FOR VISUALLY IMPAIRED PEOPLE USING NATURAL LANGUAGE PROCESSING,” *J Theor Appl Inf Technol*, vol. 31, no. 2, 2022, [Online]. Available: www.jatit.org

[14] I. Konstantinidis, M. Maragoudakis, I. Magnisalis, C. Berberidis, and V. Peristeras, “Knowledge-driven Unsupervised Skills Extraction for Graph-based Talent Matching,” in *ACM International Conference Proceeding Series*, Association for Computing Machinery, Sep. 2022. doi: 10.1145/3549737.3549769.

[15] S. Devaraju, “Natural Language Processing (NLP) in AI-Driven Recruitment Systems,” *International Journal of Scientific Research in Computer Science, Engineering and Information Technology*, pp. 555–566, Jun. 2022, doi: 10.32628/CSEIT2285241.